

ParkIt: CV-Integrated Smart Parking Allocation and Management System

PHASE – 1 REPORT

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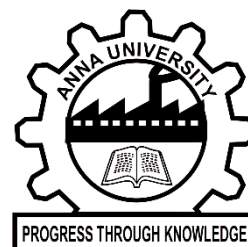
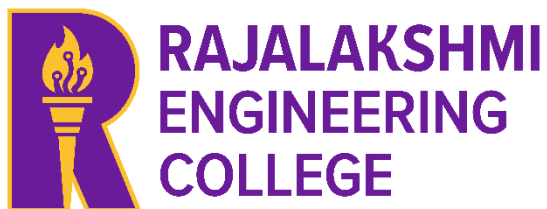
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BONAFIDE CERTIFICATE

Certified that this Report titled " **PARKIT: CV-INTEGRATED SMART PARKING ALLOCATION AND MANAGEMENT SYSTEM** " is the bonafide work of **DHARMARAJ T (230701074)**, **MANOHARAN K (230701177)**, **MITHESH THARUN S (230701184)** and **MOKESH M (230701192)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

Goal 11 – Sustainable Cities and Communities: Promotes efficient, smart, and sustainable urban parking and mobility management.

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ABSTRACT

The increasing number of vehicles and growing demand for parking facilities have created significant challenges in managing parking spaces efficiently across residential communities, educational institutions, hospitals, and commercial complexes. Conventional parking systems often rely on manual supervision and static allocation methods, resulting in increased vehicle search time, inefficient slot utilization, traffic congestion, and higher operational costs. To address these challenges, this project proposes ParkIt, a CV-Integrated Smart Parking Allocation and Management System that automates vehicle entry, intelligent parking allocation, occupancy monitoring, and exit management within a unified platform. The proposed system integrates a React Native mobile application, a web-based administrative dashboard, a Node.js backend, and a PostgreSQL database to provide real-time parking management services. Computer Vision techniques, including vehicle detection and Automatic Number Plate Recognition (ANPR), are incorporated to enable occupancy monitoring and automated vehicle verification using existing CCTV infrastructure. A smart allocation engine dynamically assigns parking slots based on real-time availability, thereby improving parking utilization and reducing user waiting time. The system aims to minimize manual intervention, enhance operational efficiency, and provide administrators with real-time insights through centralized analytics and reporting dashboards. By combining intelligent parking allocation, occupancy detection, and real-time monitoring, ParkIt offers a scalable and cost-effective solution suitable for modern smart parking environments and future smart city applications.

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TABLE OF CONTENTS

SECTION	TITLE	PAGE NO.
CHAPTER 1	INTRODUCTION	1
1.1	GENERAL	1
1.2	OBJECTIVE	2
1.3	EXISTING SYSTEM	2
1.4	PROPOSED SYSTEM	3
CHAPTER 2	LITERATURE SURVEY	5
2.1	LITERATURE REVIEW	5
2.2	COMPARISON AND DISCUSSION	7
2.3	CONCLUSION	8
CHAPTER 3	SYSTEM DESIGN	9
3.1	SYSTEM ARCHITECTURE	9
3.2	SYSTEM REQUIREMENTS	10
3.2.1	SOFTWARE REQUIREMENTS	10
3.2.2	HARDWARE REQUIREMENTS	10
3.2.3	DATASET REQUIREMENTS	11
3.2.4	DEPLOYMENT AND SCALING	11
3.3	SYSTEM DESIGN	12
3.3.1	ACTIVITY DIAGRAM	13
3.3.2	CLASS DIAGRAM	13
CHAPTER 4	PROJECT DESCRIPTION	16
4.1	METHODOLOGIES	16
4.2	MODULES	17

4.2.1	DATA MANAGER PACKAGE	19
4.2.2	CORE PACKAGE	20
4.2.3	MODERATION PACKAGE	20
4.2.4	REINFORCEMENT PACKAGE	21
4.2.5	UI PACKAGE	22
CHAPTER 5	IMPLEMENTATION AND RESULTS	23
5.1	BACKEND IMPLEMENTATION	23
5.2	COMPUTER VISION AND AI MODULE IMPLEMENTATION	25
5.3	ADMIN WEB APPLICATION IMPLEMENTATION	27
5.4	MOBILE APPLICATION IMPLEMENTATION	31
5.5	RESULTS AND PERFORMANCE EVALUATION	34
5.5.1	COMPUTER VISION AND OCCUPANCY TRACKING PERFORMANCE	34
5.5.2	OPERATIONAL EFFICIENCY AND SEARCH TIME REDUCTION	36
5.5.3	REAL-TIME SYNCHRONIZATION AND SYSTEM RESPONSIVENESS	37
5.5.4	SUMMARY OF SYSTEM PERFORMANCE METRICS	39
5.6	DISCUSSION	40
CHAPTER 6	CONCLUSION AND FUTURE WORK	41
6.1	CONCLUSION	41
6.2	FUTURE WORK	42
	REFERENCES	43

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
TABLE 1	COMPARATIVE ANALYSIS	7
TABLE 2	SOFTWARE REQUIREMENTS	10
TABLE 3	HARDWARE REQUIREMENTS	10
TABLE 5.1	OCCUPANCY DETECTION PERFORMANCE	35
TABLE 5.2	PARKING SEARCH TIME COMPARISON	37
TABLE 5.3	REAL-TIME SYNCHRONIZATION PERFORMANCE	38
TABLE 5.4	SUMMARY OF SYSTEM PERFORMANCE METRICS	39

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
FIGURE 1	OVERVIEW OF PROPOSED SYSTEM ARCHITECTURE	4
FIGURE 2	ARCHITECTURE DIAGRAM	9
FIGURE 3	ACTIVITY DIAGRAM	13
FIGURE 4	CLASS DIAGRAM	15
FIGURE 5.1	BACKEND ARCHITECTURE OF THE PARKIT SYSTEM	23
FIGURE 5.2	COMPUTER VISION PROCESSING PIPELINE	26
FIGURE 5.3	LIVE MONITORING DASHBOARD	29
FIGURE 5.4	CAMERA MONITORING INTERFACE	29
FIGURE 5.5	FACILITY CONFIGURATION SCREEN	29
FIGURE 5.6	PARKING ZONE AND SLOT CONFIGURATION INTERFACE	30
FIGURE 5.7	CAMERA CONFIGURATION SCREEN	30
FIGURE 5.8	GATE AND QR CONFIGURATION INTERFACE	30
FIGURE 5.9	HOME SCREEN	32
FIGURE 5.10	QR ACCESS SCREEN	32
FIGURE 5.11	PARKING BOOKING SCREEN	33
FIGURE 5.12	SESSION DETAILS SCREEN	33
FIGURE 5.13	USER PROFILE SCREEN	33
FIGURE 5.14	OCCUPANCY DETECTION ACCURACY	36
FIGURE 5.15	EVENT SYNCHRONIZATION LATENCY	38

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ANPR	Automated Number Plate Recognition
API	Application Programming Interface
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CV	Computer Vision
DFD	Data Flow Diagram
DQN	Deep Q-Network
EV	Electric Vehicle
FPS	Frames Per Second
GPU	Graphics Processing Unit
HD	High Definition
IoT	Internet of Things
IoU	Intersection over Union
JWT	JSON Web Token
LBP	Local Binary Patterns
LPR	License Plate Recognition
LPQ	Local Phase Quantization
MQTT	Message Queuing Telemetry Transport
OCR	Optical Character Recognition
ORM	Object-Relational Mapping
RBAC	Role-Based Access Control
Redis	Remote Dictionary Server
RFID	Radio Frequency Identification
ROI	Region of Interest
RTSP	Real-Time Streaming Protocol
SVM	Support Vector Machine
TCP	Transmission Control Protocol
UI	User Interface
WSN	Wireless Sensor Network
YOLO	You Only Look Once

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The rapid expansion of urban environments and the corresponding increase in vehicular traffic have made parking management a critical logistical challenge for modern commercial, residential, and municipal spaces. As vehicle density increases, traditional parking facilities struggle to handle high throughput efficiently, leading to localized congestion, increased carbon emissions from idling vehicles, and significant driver frustration.

The advent of artificial intelligence, specifically Computer Vision (CV) and edge computing, presents a transformative opportunity to automate and optimize these processes. The ParkIt project leverages these modern technological paradigms to create an intelligent infrastructure that manages parking operations dynamically. By shifting away from manual operations and physical hardware dependencies, ParkIt establishes a software-defined, real-time ecosystem capable of monitoring facility capacity, guiding drivers, and handling secure transactions seamlessly.

To realize this vision, the ParkIt platform seamlessly unifies Automated Number Plate Recognition (ANPR), real-time CCTV slot occupancy monitoring, dynamic slot allocation algorithms, and a cross-platform mobile application into a centralized, service-oriented architecture. By orchestrating hardware ingestion streams with micro-tier software services, the system automates vehicle entry, provides precision guidance to assigned slots, updates occupancy states dynamically without manual intervention, and executes automated session closures upon exit. Ultimately, this comprehensive integration bridges the gap between facility administrators and commuters, establishing a scalable and cost-effective blueprint for modern urban mobility.

1.2 OBJECTIVE

The primary objective of the ParkIt system is to engineer an end-to-end intelligent parking management and facility automation platform designed to replace labor-intensive operations with a computer-vision-driven infrastructure.

Specifically, the system is designed to achieve the following operational metrics:

- **Optimize Vehicle Routing:** To reduce the average parking search time from the traditional 3 to 10 minutes down to under 60 seconds via pre-assigned dynamic routing algorithms.
- **Digitize Access Control:** To entirely eliminate physical paper ticketing through the implementation of digital QR verification and automated license plate validation.
- **Ensure High-Accuracy Monitoring:** To maintain a real-time slot occupancy tracking accuracy exceeding 95%, ensuring robust performance across varied lighting and visual occlusion conditions.
- **Centralize Facility Management:** To achieve the complete integration of security protocols, billing and logging operations, and facility analytics into a single centralized platform.

1.3 EXISTING SYSTEM

Current parking management solutions in modern commercial and municipal environments suffer from high operational friction and systemic inefficiencies. The existing systems rely heavily on traditional, labor-intensive operational models that utilize manual allocation processes and physical paper ticketing at entry booths.

A major flaw in the existing infrastructure is the fundamental lack of real-time visibility regarding available spaces, which directly causes localized traffic bottlenecks and poor overall slot utilization as drivers aimlessly circle facilities searching for vacancies. Furthermore, when automated occupancy tracking is attempted in existing systems, it typically relies on physical ground sensors installed in every individual parking slot. These traditional hardware sensors are expensive to deploy at scale, difficult to maintain, and prone to physical damage. Current systems also fail to unify the driver experience, often separating the entry mechanism, navigation, and billing into disjointed, manual steps.

1.4 PROPOSED SYSTEM

To comprehensively address the inefficiencies of current models, the proposed ParkIt system is designed to automate operations by unifying Automated Number Plate Recognition (ANPR), real-time CCTV slot occupancy detection, and dynamic slot allocation algorithms.

The proposed system replaces expensive physical ground sensors with a centralized Computer Vision Layer. It utilizes standard overhead CCTV cameras streaming video via RTSP to a standalone Python processing service. This service executes a multi-stage pipeline using custom-trained YOLOv8 models to detect vehicles within mapped 2D polygon coordinate spaces, effectively creating a real-time digital twin of the physical parking facility.

Furthermore, the proposed system introduces a decoupled, service-oriented architecture that bridges the gap between administrators and drivers. It features a Flutter cross-platform mobile application that provides commuters with dynamic QR generation, real-time slot status updates, and turn-by-turn visual prompts guiding them directly to their assigned slot. Concurrently, a web-based administrative dashboard offers facility managers a live interactive floor plan,

real-time analytics, and dynamic camera configuration tools. By orchestrating these hardware ingestion streams with micro-tier software services, the proposed system seamlessly automates vehicle entry, provides precision guidance, updates occupancy dynamically without manual intervention, and executes automated session closures upon exit.

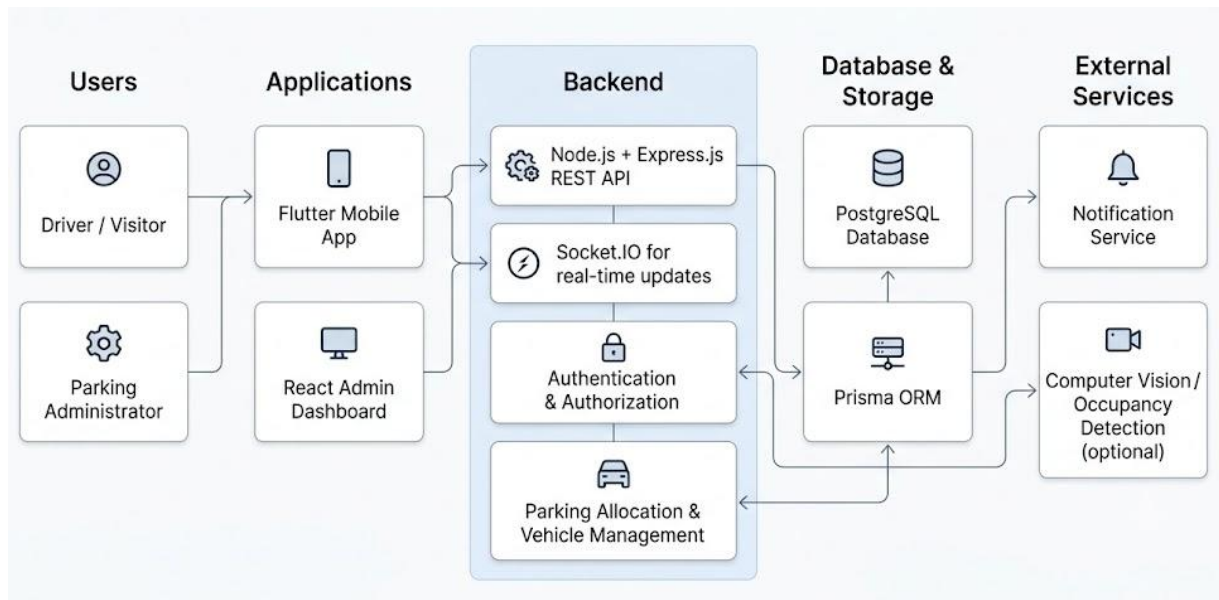


Fig 1. Overview of Proposed System Architecture

As illustrated in Figure 1, the proposed system features a decoupled, service-oriented architecture that connects drivers using a Flutter mobile application and administrators using a React web dashboard to a central Node.js and Express.js backend. This backend executes core business operations—including secure authentication, dynamic slot allocation, and vehicle management—while broadcasting live facility updates to clients via Socket.IO. All persistent data is managed in a PostgreSQL database using Prisma ORM, while the core application layer seamlessly integrates with external notification services and a Computer Vision processing engine to deliver real-time slot occupancy monitoring and facility automation.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

The landscape of smart parking management has experienced a profound evolution, transitioning from manual, static administrative models to highly automated, data-driven ecosystems that integrate proximity IoT sensing, advanced computer vision, and machine learning algorithms. Early research efforts heavily emphasized hardware-based proximity sensing and microcontrollers to monitor facility capacity. For instance, Elsonbaty and Shams (2020) and Pham et al. (2015) designed indoor smart parking systems utilizing Arduino Uno, NodeMCU, ESP8266 Wi-Fi modules, and infrared obstacle sensors alongside piezoelectric mats to detect vehicle presence and log occupancy metrics into centralized relational databases. Similarly, Pala and Inan (2007) explored active and passive RFID technologies deployed via USB and Ethernet-connected readers at entry and exit gates, successfully enabling rapid, unmanned check-outs without requiring vehicles to come to a complete stop, thereby reducing traffic queues and idling emissions. To optimize power consumption and eliminate heavy ground excavation, Chinrungrueng et al. (2007) introduced an optical wireless sensor network (WSN) featuring a star topology and polling MAC protocol, equipping sensor nodes with dual optical sensor heads to accurately count vehicles and track travel direction at garage boundaries.

In parallel with physical sensor grids, computer vision and image processing methodologies have emerged as powerful alternatives for wide-area monitoring and license plate recognition. López Sastre et al. (2007) pioneered symbolic 2D homography algorithms using algebraic homomorphism and Gabor filter bank texture extractions combined with Support Vector Machines (SVM) to classify parking conditions from a single camera view, achieving a 99.73% classification accuracy under varying shadow and illumination conditions. Addressing license

plate recognition (LPR) in complex environments, Weihong and Jiaoyang (2020) evaluated deep learning detection frameworks, noting that YOLO-based localization combined with segmentation-free sequence labeling—such as LSTM networks using Connectionist Temporal Classification (CTC)—effectively avoids character segmentation errors. Selmi et al. (2017) similarly developed a two-stage CNN system for plate classification and character OCR, achieving high precision across benchmark datasets. Furthermore, Almeida et al. (2015) established a standard for visual classification by introducing the PKLot dataset containing nearly 700,000 images, evaluating Local Binary Patterns (LBP) and Local Phase Quantization (LPQ) texture descriptors paired with SVM classifiers.

Beyond presence detection, machine learning and predictive modeling have become critical for anticipating future availability and vehicle dwell times. Barker and Rehman (2019) evaluated five machine learning algorithms—including K-Nearest Neighbor, KStar, and Random Forest—using the WEKA workbench to predict parking occupancy rates under varying traffic schedules. Canli and Toklu (2021) advanced mobile application design by developing the ISTPARK system, which incorporated a Keras-trained LSTM neural network analyzing multivariate time-series data from over 4.5 million records, achieving a peak prediction accuracy of 99.57%. In a similar vein, Shalini M.K. et al. (2025) demonstrated that Random Forest models can accurately anticipate parking duration with high precision, while clustering algorithms optimize slot assignments based on floor levels and vehicle dimensions. Finally, advanced system optimization and routing algorithms have focused on minimizing user search times and maximizing facility throughput. Pranaav S. and Seshasayee (2025) integrated modified Dijkstra's algorithms with ANPR and AR-guided navigation, reducing average parking seek times significantly. Extending optimization further, Yuan et al. (2023) formulated a multi-objective shared parking allocation model solved via an improved Deep

Q-Network (DQN) reinforcement learning algorithm, outperforming linear weighting models by over 12% and drastically cutting user walking distances.

2.2 COMPARISON AND DISCUSSION

A rigorous comparative analysis of the reviewed literature reveals distinct architectural trade-offs across hardware-dense sensor grids, pure computer vision systems, and hybrid software platforms.

Evaluation Dimension	Proximity & IoT Sensing (WSN/RFID)	Pure Computer Vision & LPR	Hybrid & Deep Learning Platforms
Core Technologies	Arduino, NodeMCU, Ultrasonic, IR, RFID tags, WSN nodes.	OpenCV, YOLOv8, CNN, Gabor filters, LBP/LPQ texture descriptors.	YOLOv8 object detection, LSTM time-series models, DQN reinforcement learning.
Primary Advantages	Highly reliable local presence detection; low latency for simple binary states.	Wide-area coverage from overhead cameras; eliminates expensive per-slot sensor wiring.	Dual-verification eliminates false positives; superior prediction and dynamic allocation.
Critical Limitations	High installation, excavation, and battery maintenance costs; vulnerable to node failure.	Highly sensitive to environmental degradation, weather noise, night-time glare, and occlusions.	Higher computational overhead on edge hardware; complex multi-model pipeline integration.
Peak Performance	Fast gate check-ins and reliable count verification.	96% to 99% classification and OCR accuracy under stable controlled lighting.	Over 99% prediction accuracy (LSTM) and sub-0.2s optimization response times (DQN).

Table 1 Comparative Analysis

The discussion of these findings indicates that while proximity sensors and RFID networks provide reliable binary presence detection, their deployment across large urban facilities is economically prohibitive due to high infrastructure and maintenance overheads. Conversely, pure computer vision frameworks offer broad scalability through overhead CCTV cameras, yet they suffer from severe environmental fragility. Factors such as heavy rain, dense fog, low-light night

conditions, and transient visual occlusions—like a car blocked by an adjacent high-profile vehicle or tree shadows—frequently trigger false negatives in occupancy and vacancy tracking. Furthermore, classifiers exhibit a noticeable generalization deficit, experiencing sharp accuracy drops (such as drops from 99% to 89%) when transferred to unseen environments with differing camera mounting angles or ground textures. Consequently, current research highlights critical gaps in achieving viewpoint invariance, robust low-light video processing, and energy-efficient edge computation, establishing a clear mandate for hybrid architectures that combine the fault-tolerance of physical boundary counters with the panoramic flexibility of computer vision.

2.3 CONCLUSION

The comprehensive review of existing literature establishes a robust technical and theoretical foundation for designing advanced, context-aware smart parking management systems. By evaluating past methodologies—ranging from optical wireless sensor networks and RFID gate controls to deep learning LPR and reinforcement learning allocation engines—the survey provides actionable blueprints for resolving modern urban parking inefficiencies. Specifically, adopting dual-head optical sensing concepts for entry and exit vehicle direction tracking, combined with homography-based perspective transformations and texture feature descriptors, offers a highly effective pathway to secure reliable slot occupancy validation without the prohibitive costs of per-slot physical sensor grids. Furthermore, integrating deep learning object detection with dynamic routing algorithms ensures that future parking platforms can minimize search times, eliminate physical paper ticketing, and deliver seamless, automated user experiences.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The ParkIt platform utilizes a decoupled, service-oriented architecture designed to simultaneously process real-time data streaming, computer vision inference, and standard transactional database operations. The architecture is structured into four primary layers that ensure clear separation of concerns and high scalability. The Client Layer comprises a cross-platform mobile application built using React Native and Expo SDK for end-users, alongside a web-based administrative dashboard developed with React.js, Vite, and TypeScript. The API Gateway and Application Layer features a Node.js and Express.js server that handles RESTful API endpoints for transactional operations alongside a Socket.IO WebSocket server engine responsible for pushing real-time slot state changes and alerts to connected frontend clients. The Computer Vision Layer operates as an asynchronous Python service that ingests high-definition RTSP video feeds from facility CCTV cameras, executing OpenCV frame transformations, custom-trained YOLOv8 models for occupancy detection.

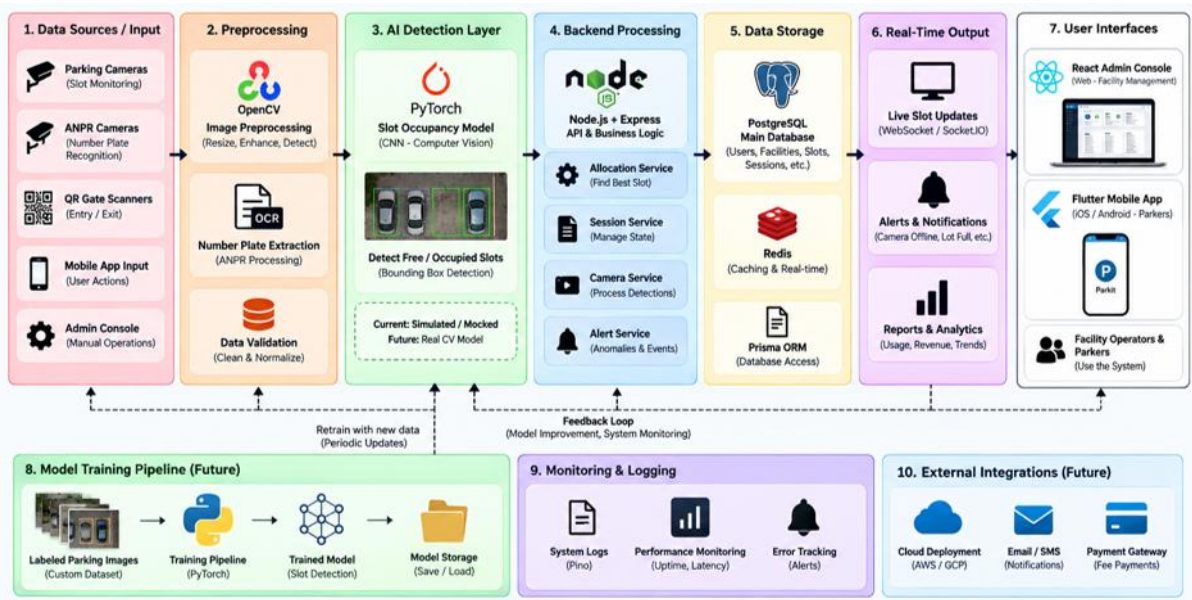


Fig 2. Architecture Diagram

3.2 SYSTEM REQUIREMENTS

The execution and deployment of the ParkIt platform require specific software environments, processing hardware, and containerized dependencies to support real-time computer vision inference and concurrent web traffic.

3.2.1 SOFTWARE REQUIREMENTS

Software Component	Technology / Tool	Version / Specification
Operating System	Ubuntu Linux / Windows (WSL2)	Ubuntu 22.04 LTS recommended for production
Backend Runtime	Node.js / Express.js	Node.js v18.x+, Express.js
CV Engine Runtime	Python	Python 3.10+
Frontend Frameworks	React.js, React Native, Vite	React 18, Vite, Expo SDK
Database & ORM	PostgreSQL, Prisma ORM	PostgreSQL 15, Prisma Client
Caching & Messaging	Redis	Redis 7 (Pub/Sub & State Caching)
AI & Vision Libraries	PyTorch, OpenCV, PaddleOCR	YOLOv8 weights (.pt), OpenCV, PaddleOCR
DevOps & Containerization	Docker, Nginx, GitHub Actions	Docker Compose, NVIDIA Container Toolkit

Table 2 Software Requirements

3.2.2 HARDWARE REQUIREMENTS

Hardware Category	Specification / Component	Purpose in System
Processor (CPU)	Multi-core x86_64 (Intel i7 / Ryzen 7 or higher)	Powers the API gateway, web servers, and request routing
Graphics (GPU)	NVIDIA GPU with CUDA support (Min. 6GB VRAM)	Accelerates YOLOv8 object detection and ANPR extraction
System Memory	Minimum 16 GB RAM	Handles multi-threaded RTSP stream buffering and caching
Camera Ingestion	HD/4K CCTV Cameras with RTSP support	Captures live video feeds for the Python computer vision engine
Gate Hardware	Optical sensors and hardware relay controllers	Detects vehicle arrival and operates physical entry/exit barriers

Table 3 Hardware Requirements

3.2.3 DATASET REQUIREMENTS

The computer vision and artificial intelligence components of the ParkIt platform rely on carefully curated and pre-trained datasets to achieve high precision in vehicle detection, slot occupancy tracking, and license plate recognition. Specifically, the object detection pipeline utilizes YOLOv8 Nano and Small models that are initially pre-trained on the comprehensive COCO dataset for general object classification and subsequently fine-tuned on custom-curated parking lot image datasets. These custom datasets encompass diverse spatial angles, lighting conditions, and perspective scales to ensure accurate region-of-interest (ROI) polygon mapping. Additionally, the Automated Number Plate Recognition (ANPR) engine requires specialized license plate image datasets featuring varied regional alphanumeric formats, fonts, and environmental backgrounds to optimize optical character recognition (OCR) and text extraction performance under mixed weather scenarios.

3.2.4 DEPLOYMENT AND SCALING

The deployment and infrastructure architecture of ParkIt is designed to orchestrate standard containerized web services alongside GPU-accelerated computer vision workloads seamlessly. The entire platform is containerized using Docker and managed via a centralized docker-compose.yml configuration file that establishes isolated network bridges and persistent volume mounts for services including PostgreSQL, Redis, the Express gateway backend, and the Python CV engine. Horizontal and vertical scaling is achieved by leveraging the NVIDIA Container Toolkit to allocate dedicated GPU resources (count: 1, capabilities: [gpu]) to the Python processing container, ensuring high-throughput RTSP video frame inference. Meanwhile, the backend gateway and web services can be horizontally scaled behind an Nginx reverse proxy to handle high volumes of concurrent mobile user traffic and real-time WebSocket connections.

3.3 SYSTEM DESIGN

The system design of the ParkIt platform bridges high-level architectural abstractions with concrete technical execution, defining the transactional boundaries, state machines, and messaging pipelines that govern facility automation. The design prioritizes low latency and fault tolerance by decoupling real-time computer vision inference from standard database transactions. State transitions—such as a parking slot shifting from vacant to occupied—are managed through an asynchronous event-driven model powered by a Redis message queue and pub/sub broker. This ensures that high-frequency video frame processing does not block critical database operations or API response times for mobile users. Furthermore, concurrency control mechanisms, such as Redis-based transactional locks during slot reservation, eliminate race conditions when multiple vehicles arrive simultaneously.

3.3.1 ACTIVITY DIAGRAM

The activity workflow models the end-to-end execution path of a vehicle and the corresponding system reactions across hardware and software components. Upon vehicle arrival at the facility perimeter, an optical sensor or loop detector triggers the ANPR CCTV camera to capture a high-resolution frame. The Python computer vision engine processes this frame using a lightweight YOLOv8 model to localize the license plate, followed by PaddleOCR extraction and regional regex validation. Simultaneously, the gateway checks the PostgreSQL database via Prisma ORM to match the license plate against active user accounts or generate an anonymous visitor session. Once verified, the smart parking allocation engine computes the optimal vacant slot by evaluating spatial distance, vehicle class alignment, and traffic density. A Redis transactional lock secures the chosen slot, a new parking session record with an ACTIVE status is instantiated, and the assigned slot details are pushed via Socket.IO to the driver's mobile application. During the exit sequence, a similar workflow triggers dwell-time

calculation, automated wallet debit or terminal payment processing, session closure, and computer vision slot-clearance verification before opening the physical barrier.

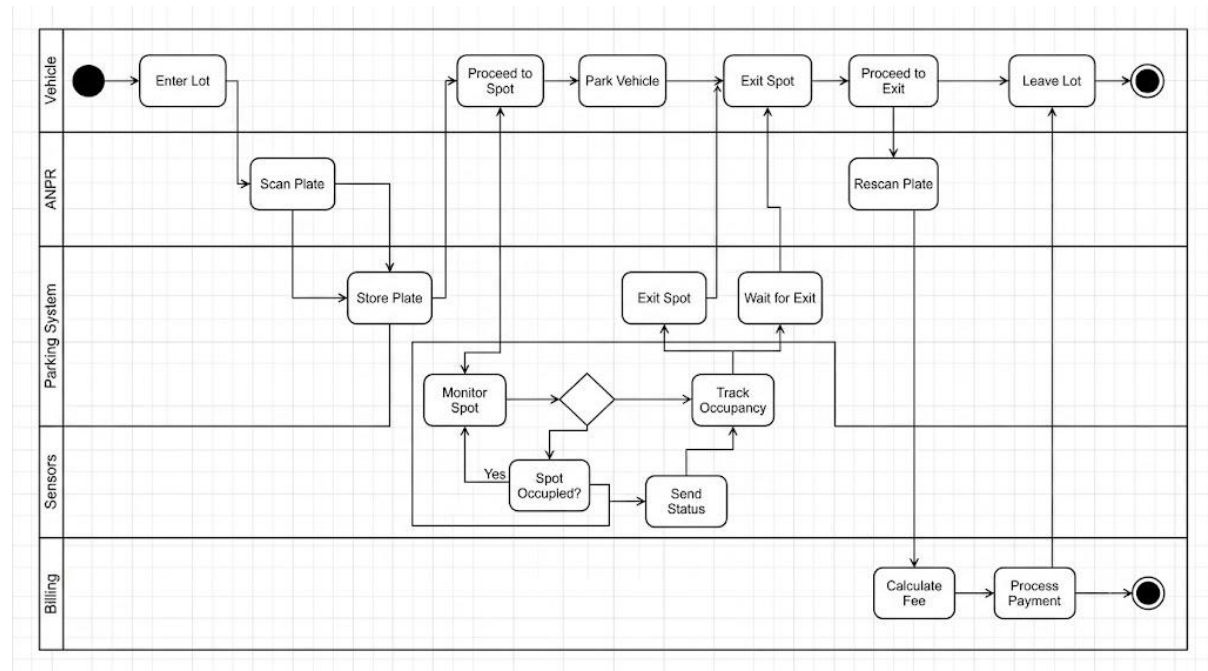


Fig 3. Activity Diagram

As illustrated in Fig 3, the activity diagram maps out the sequential execution flow of the system, detailing the operational logic from the moment a vehicle arrives at the facility gate and triggers the optical sensor, through the ANPR scanning and QR validation steps, down to the automated session initialization, dynamic slot allocation computation, real-time WebSocket broadcasting, and final exit billing reconciliation.

3.3.2 CLASS DIAGRAM

The class diagram represents the structural organization of the ParkIt system by defining its major entities, attributes, and relationships. It provides a clear view of how the facility, parking infrastructure, users, vehicles, monitoring components, and parking transactions are connected within the system. The design is based on the data requirements of the backend and supports the management of parking operations across different facilities and zones.

The Facility class represents a parking-enabled location and maintains basic information such as facility name, address, contact details, total parking capacity, and active status. Each facility can contain multiple Zone, ParkingSlot, Camera, Gate, AdminUser, ParkingSession, and Alert records. The Zone class is used to organize parking spaces based on floors or designated areas, while the ParkingSlot class maintains individual slot information such as slot code, type, status, coordinates, and distance from the gate. The Camera class represents the surveillance cameras configured within a facility. Cameras may be associated with a particular zone and can monitor one or more parking slots. The CameraSlotCoverage class maintains this many-to-many coverage relationship and identifies whether a camera is the primary camera for a particular slot and whether the coverage has been confirmed by an administrator. The Gate class represents entry and exit points and can optionally be associated with a camera for vehicle verification and monitoring. The user-related portion of the model consists of the User, Vehicle, and ParkingSession classes. A user can register multiple vehicles and can create multiple parking sessions. Each vehicle can participate in multiple parking sessions over time. A parking session connects the user and vehicle with a specific facility and parking slot while recording allocation time, entry and exit times, parking duration, fee, payment status, and the current session status. Entry and exit gates are also associated with the session to maintain the complete movement history of a vehicle within the facility. The Detection class stores observations generated by the camera monitoring system. Each detection can be associated with a camera, parking slot, or gate and records information such as detection type, detected status, detected license plate, confidence score, bounding box, and detection time. These records provide the basis for occupancy monitoring and vehicle identification within the system.

The AdminUser class represents authorized personnel responsible for managing a facility. Administrators can manage facility configurations and resolve system-generated alerts. The Alert class records operational issues such as camera problems, parking-slot conditions, gate events, or session-related events. Each alert contains a severity level, message, resolution status, and references to the administrator who resolved it. The relationships shown in the class diagram ensure that the system maintains consistency between physical parking infrastructure and operational records. The model also supports historical tracking because vehicles, detections, parking sessions, and alerts can be associated with multiple events over time. This structure provides a foundation for implementing the ParkIt backend using a relational database while allowing the administrative and mobile applications to access the required information through the system APIs.

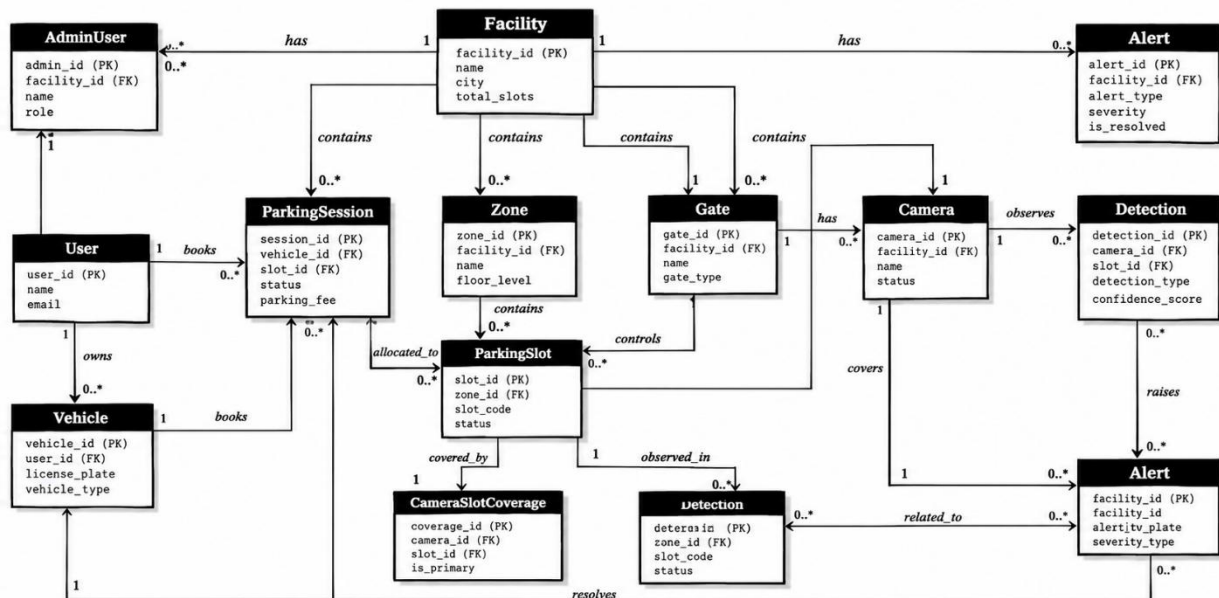


Fig 4. Class Diagram

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The ParkIt platform implements a robust, service-oriented methodological framework designed to bridge hardware ingestion streams with modern cloud-native software engineering practices. The core methodology revolves around asynchronous event-driven processing, separating high-frequency computer vision operations from standard relational database transactions to maintain sub-second responsiveness.

- **Computer Vision and Inference Pipeline:** The system utilizes multi-threaded OpenCV video readers to ingest high-definition RTSP streams from overhead facility cameras at a controlled sampling rate of 3 to 5 frames per second. Incoming frames are processed via a fine-tuned YOLOv8 model combined with PaddleOCR for Automated Number Plate Recognition (ANPR) and spatial occupancy classification. To eliminate false positives caused by temporary occlusions, glare, or passing pedestrians, the methodology enforces a sliding-window state debouncing protocol across consecutive frames.
- **Spatial Mapping and IoU Calculation:** Physical parking slots are mapped into the system using custom 2D polygon coordinates. Vehicle occupancy is determined by evaluating the Intersection-over-Union (IoU) ratio between a detected vehicle's bounding box and the slot's polygon, triggering an occupancy state transition when the threshold exceeds 0.45.
- **Dynamic Weighted Allocation Methodology:** Rather than relying on random or sequential space assignment, ParkIt employs a weighted optimization algorithm during slot calculation. This methodology evaluates multi-variable criteria, including shortest-path spatial distance

from entry gates, vehicle class dimensions (sedan, SUV, EV, accessible), facility traffic load balancing across aisles, and priority reservations for subscription or VIP holders.

- **Real-Time Synchronization and Concurrency Control:** System state changes are published via a Redis in-memory event bus and pub/sub message queue. Redis-based transactional locks are implemented during slot reservations to prevent race conditions when multiple vehicles enter simultaneously. Relational persistence and type-safe data integrity are maintained using PostgreSQL coupled with Prisma ORM.

4.2 MODULES

The system functionality is segmented into six specialized, highly cohesive modules that govern the complete lifecycle of a parking session:

Module 1: Entry Management System

- **Core Responsibility:** Validates incoming vehicle credentials, captures arrival metrics, generates temporary access tokens, and initializes tracking logs.
- **Functional Mechanics:** Integrates hardware gate barrier controllers (via MQTT/REST relay switches) with automated ANPR cameras and dynamic mobile QR code verification. It performs rigorous validation checks, including registration status, blacklisted vehicle screening, capacity limits, and anti-pass back rule enforcement.

Module 2: Smart Parking Allocation Engine

- **Core Responsibility:** Computes and assigns the optimal parking space using contextual criteria and traffic balancing logic.
- **Functional Mechanics:** Executes weighted path optimization algorithms that factor in spatial distance, vehicle class alignment, floor congestion, and

VIP/subscription priorities. Utilizes Redis concurrency locks to ensure safe, deadlock-free slot assignments.

Module 3: Computer Vision & Occupancy Monitoring

- Core Responsibility: Maintains a real-time digital twin of physical parking space occupancy without requiring physical ground sensor hardware.
- Functional Mechanics: Processes multi-camera RTSP feeds simultaneously, projecting 3D bounding boxes into 2D polygon coordinate spaces defined in the database. Operates with real-time target frame processing at ≥ 15 FPS per stream to ensure sub-second state reflection.

Module 4: Exit Management & Billing System

- Core Responsibility: Reconciles active parking sessions, calculates fees based on exact dwell times, processes payments, and triggers exit barrier releases.
- Functional Mechanics: Connects exit ANPR camera output to session matching services, applying facility rate rules, subscription benefits, and grace periods. Integrates payment gateway APIs for automated wallet debits while offering manual exit booth ticket validation.

Module 5: Administrative Management Dashboard

- Core Responsibility: Provides facility operators with a centralized interface for live facility monitoring, video feed management, and business intelligence reporting.
- Functional Mechanics: Built using a Claymorphism UI design system with tactile elements and high-contrast status colors. Features an interactive multi-floor graphical layout for manual state overrides, a dynamic camera configuration tool for drawing ROI polygon masks over live video streams, and real-time usage analytics.

Module 6: Mobile User Experience Application

- Core Responsibility: Empowers end-user drivers with tools to navigate, verify entry, locate assigned slots, and settle digital payments.
- Functional Mechanics: Developed using React Native and Expo SDK, incorporating vector floor map rendering, dynamic time-based OTP QR generation for secure entry, and turn-by-turn visual navigation routing drivers directly to their designated bay.

4.2.1 DATA MANAGER PACKAGE

The Data Manager Package serves as the relational persistence and data abstraction layer of the ParkIt platform, architected using PostgreSQL and managed via Prisma ORM. This package is responsible for enforcing rigorous data integrity, type-safe query execution, and performance-optimized indexing across all core domain entities, including user accounts, registered vehicles, physical facilities, multi-floor layouts, individual parking slots, CCTV cameras, active parking sessions, and entry/exit audit logs. By utilizing Prisma's schema definition language, the data manager orchestrates complex relational bindings and cascade deletion protocols—such as cascading deletions from facilities down to floors and slots, or handling set-null constraints on unassigned vehicle sessions—thereby preventing data fragmentation and ensuring absolute database consistency. Furthermore, the package implements strategic indexing on high-frequency lookup columns, including license plate strings, slot status states, and composite facility-status keys, which guarantees sub-millisecond query performance during rapid vehicle ingress/egress transactions and real-time computer vision state updates.

4.2.2 CORE PACKAGE

The Core Package encapsulates the primary business logic, RESTful API controllers, and real-time event dispatch engines of the backend gateway, developed on a Node.js and Express.js runtime. This package governs the essential operational workflows of the system, housing controllers for user authentication, vehicle management, and session initialization, while enforcing role-based access control (RBAC) via secure JWT middleware. At the heart of the core package is the Smart Parking Allocation Engine, which executes advanced weighted optimization algorithms to compute optimal slot assignments by evaluating spatial distance, vehicle class compatibility, and facility traffic density. Additionally, the core package integrates a Socket.IO WebSocket server engine and a Redis pub/sub message broker, facilitating instant bi-directional event broadcasting—such as *slot:status_changed* and *session:created* alerts—to connected mobile applications and administrative dashboards while managing transactional locks to eliminate concurrency race conditions.

4.2.3 MODERATION PACKAGE

The Moderation Package governs security oversight, exception handling, and administrative intervention workflows across physical entry gates and digital management terminals. This package provides security guards and facility administrators with specialized tools to review, override, and resolve operational exceptions, such as unreadable license plates caused by adverse weather, dust accumulation, or physical occlusion, as well as unauthorized or blacklisted access attempts. When an Automated Number Plate Recognition (ANPR) scan experiences a confidence failure below acceptable operational thresholds or results in an explicit mismatch against active database records, the moderation package automatically diverts the event into a secure manual validation queue. Within this interface, authorized personnel can inspect captured vehicle snapshots, execute manual license plate string corrections, or trigger physical gate

relay overrides via MQTT and REST integration. Furthermore, this package enforces strict role-based access control (RBAC) policies across all interface levels, ensuring that security guards and facility admins operate strictly within their designated operational scopes while maintaining comprehensive audit logs of all manual interventions, gate overrides, and system configurations for complete institutional accountability.

4.2.4 REINFORCEMENT PACKAGE

The Reinforcement Package encapsulates the advanced optimization, dynamic routing, and load-balancing algorithms that dictate intelligent space allocation and traffic distribution across multi-floor parking facilities. Moving beyond random or static slot assignment methods, this package evaluates multi-variable contextual criteria to compute the optimal parking space for every incoming vehicle. The mathematical allocation logic is governed by a multi-criteria optimization function that computes an overall suitability score S_{slot} for each available candidate slot based on weighted operational parameters:

$$S_{slot} = w_1 \cdot \text{Dist}(g, s) + w_2 \cdot \text{Match}(v, s) + w_3 \cdot \text{Load}(a)$$

where $\text{Dist}(g, s)$ represents the shortest-path spatial trajectory from entry gate g to candidate slot s , $\text{Match}(v, s)$ denotes the compatibility score between vehicle class specifications v (such as sedans, SUVs, electric vehicles, or accessible bays) and slot physical dimensions, and $\text{Load}(a)$ measures the current traffic density and congestion index within aisle a , scaled by respective weighting coefficients w_1, w_2, w_3 . To maintain spatial accuracy during automated slot tracking, the package coordinates with the computer vision layer's Intersection-over-Union (IoU) validation threshold which ensures that vehicle presence within 2D region-of-interest polygons is reliably verified before triggering state transitions.

4.2.5 UI PACKAGE

The UI Package provides the user-facing interfaces of the ParkIt system and connects the application screens with the backend services. It consists of two major applications: the web-based administrative interface and the mobile application used by parking users. The interfaces are designed to present parking information in a clear and responsive manner while allowing users and administrators to perform their respective operations efficiently. The Admin Web Application is developed using React.js and provides administrators with centralized control over facility configuration and parking operations. It includes interfaces for managing facilities, parking zones, parking slots, cameras, gates, and QR-based access settings. The dashboard displays the current parking status and monitoring information, enabling administrators to observe occupancy and system events from a single interface. Configuration screens allow authorized administrators to update infrastructure details without directly modifying the underlying database. The Mobile Application is developed using Flutter and provides parking users with access to the main parking services. Users can view available parking information, access QR-based entry functionality, make parking bookings, and view active or previous parking sessions. The application also provides profile and vehicle-related information required for managing parking activities. The UI Package communicates with the backend through REST APIs and real-time communication services. Data received from the backend is presented dynamically on the respective screens, allowing changes in parking availability and session status to be reflected without requiring manual database interaction. This separation between the user interfaces and backend services improves maintainability and allows the administrative and mobile applications to evolve independently while using the same core system services.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 BACKEND IMPLEMENTATION

The backend serves as the central processing unit of the ParkIt system, enabling seamless communication between the mobile application, administrative dashboard, database, and intelligent parking management services. It is responsible for handling user requests, managing parking sessions, allocating parking slots, processing gate operations, and maintaining real-time synchronization across the entire platform. The backend is developed using Node.js and Express.js with TypeScript support to ensure scalability, maintainability, and efficient request handling. A REST based architecture is adopted to facilitate communication between client applications and server-side services. This architecture allows different modules of the system to operate independently while maintaining a structured flow of data throughout the application.

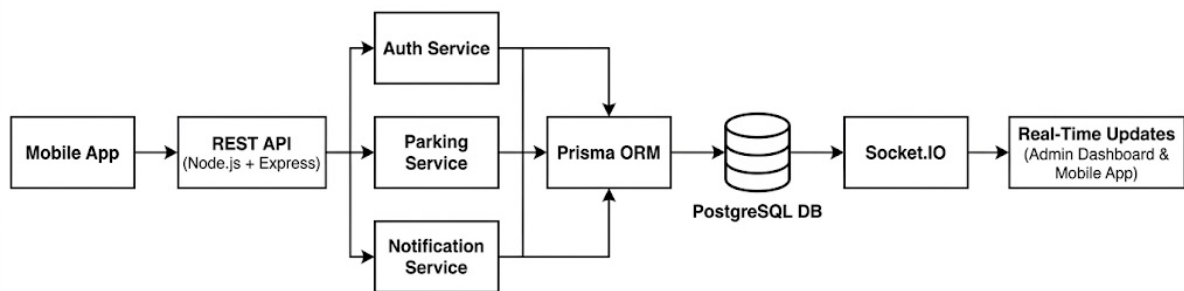


Fig 5.1 Backend Architecture of the ParkIt System

To manage persistent data, PostgreSQL is used as the primary relational database, while Prisma ORM provides a simplified and type-safe interface for database operations. The database stores information related to facilities, parking zones, parking slots, cameras, gates, users, vehicles, parking sessions, and system alerts. This structured storage mechanism enables efficient retrieval and management of parking information while maintaining data consistency.

The backend is organized into multiple service modules, each dedicated to a specific functional responsibility. The authentication service manages user registration, login, and token validation. The allocation engine identifies and assigns the most suitable parking slot based on predefined allocation criteria. Session management services handle the complete lifecycle of a parking session, including vehicle entry, parking confirmation, occupancy updates, and vehicle exit processing. Additional services support facility management, onboarding operations, alert generation, reporting functions, and background maintenance tasks.

Security is implemented through JSON Web Token (JWT) based authentication and role-based access control mechanisms. These features ensure that only authorized users can access system resources according to their designated roles. Administrative users can manage parking facilities and monitor operations, while mobile application users can access only their own parking sessions and vehicle information. This approach enhances system security and protects sensitive parking data from unauthorized access.

To support live parking monitoring, Socket.IO is integrated into the backend architecture. Whenever a parking slot changes status or a parking session is updated, real-time events are broadcast to connected administrative dashboards. This enables administrators to instantly view occupancy changes, vehicle movements, and system alerts without manually refreshing the interface. Such real-time communication improves operational awareness and facilitates efficient parking management.

The backend also incorporates validation, error handling, and logging mechanisms to improve reliability and maintainability. Incoming requests are validated before processing, and meaningful responses are returned in the event of invalid inputs or operational failures. Logging services assist in monitoring system activity and support troubleshooting during deployment and maintenance.

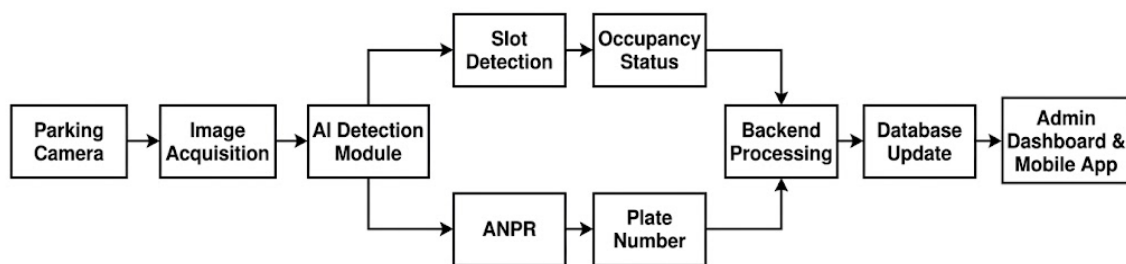
Overall, the backend implementation provides a robust foundation for the ParkIt platform by ensuring secure communication, efficient data management, real-time synchronization, and reliable parking operations across the administrative and mobile environments.

5.2 COMPUTER VISION AND AI MODULE IMPLEMENTATION

The Computer Vision and Artificial Intelligence module forms the intelligent component of the ParkIt system. Its primary objective is to automate parking space monitoring and vehicle identification using camera-based analysis. The module is designed to support parking slot detection, occupancy monitoring, and Automatic Number Plate Recognition (ANPR), thereby reducing manual intervention and improving operational efficiency. Although the current implementation utilizes a simulated AI adapter for development and testing purposes, the system architecture has been designed to seamlessly integrate real-time computer vision models in future deployments.

During facility onboarding, parking slot cameras capture reference images of the parking area when all slots are vacant. These images are used to identify individual parking spaces through an AI-assisted extraction process. The detected parking regions are represented using bounding boxes and presented to administrators for verification. Confirmed detections are converted into parking slot records within the system database, while duplicate or incorrect detections can be manually rejected. This semi-automated process simplifies parking lot configuration and reduces setup effort for facility operators. The vehicle identification component is implemented through an Automatic Number Plate Recognition workflow. When a driver scans the QR code located at an entry or exit gate, the corresponding gate camera captures the vehicle image and extracts the license plate information. The detected vehicle number is then used to associate the vehicle with a parking session and facilitate parking slot allocation

or session completion. This process enables a contactless and efficient vehicle management experience while maintaining accurate vehicle tracking records. To ensure modularity and future scalability, ParkIt employs an AI Adapter architecture. All computer vision functionalities are abstracted through a common interface, allowing different AI models to be integrated without modifying the core business logic of the application. The current mock implementation generates simulated parking slot detections and vehicle plate readings for development and testing purposes. When a production-ready model becomes



available, the mock adapter can be replaced with a real AI implementation with minimal system changes.

Fig 5.2 Computer Vision Processing Pipeline

The future version of the system is planned to integrate deep learning-based object detection and optical character recognition techniques. Parking slot occupancy detection will continuously analyze video frames captured by parking cameras to determine whether a slot is occupied or available. Similarly, ANPR models will automatically identify vehicle registration numbers from gate cameras. The processed results will be transmitted to the backend server, where parking slot statuses and vehicle sessions will be updated in real time. The AI module also supports future enhancements such as intelligent parking allocation, occupancy prediction, and anomaly detection. These capabilities can help optimize parking utilization, improve user convenience, and generate automated alerts for unusual parking activities or system failures. Through its modular architecture, the Computer Vision and AI module provides a strong foundation

for transforming conventional parking facilities into intelligent and data-driven parking environments.

5.3 Admin Web Application Implementation

The Admin Web Application serves as the operational control center of the ParkIt system. It enables administrators to manage parking facilities, monitor parking occupancy, configure cameras and gates, oversee parking sessions, and respond to system-generated alerts through a centralized web interface. The application is designed to provide real-time visibility into parking operations while simplifying administrative tasks through an intuitive and responsive dashboard. The administrative portal is developed using React.js and Vite, providing a modern and interactive user experience. The frontend communicates with backend services through REST APIs and receives real-time updates through Socket.IO connections. This architecture ensures that administrators can view the latest parking information without manually refreshing the application. As parking slots change occupancy status or vehicles enter and exit the facility, the dashboard automatically reflects the updated information.

The dashboard acts as the primary interface for monitoring parking operations. It presents key information such as total parking slots, occupied slots, available slots, active parking sessions, recent vehicle movements, and system alerts. This consolidated view allows administrators to quickly assess facility utilization and identify operational issues requiring immediate attention. Facility management functionalities enable administrators to create and configure parking facilities, define parking zones, register gates, and manage camera installations. During facility onboarding, administrators can upload parking lot images, verify AI-generated parking slot detections, and confirm parking slot configurations before activating the facility. This process ensures that parking layouts accurately represent real-world conditions and support efficient parking allocation.

The parking session management module allows administrators to track vehicle movements throughout the parking lifecycle. Detailed information regarding vehicle entries, allocated parking slots, parking durations, and completed exits can be accessed through dedicated monitoring screens. This functionality improves transparency and enables administrators to investigate parking-related issues whenever required. The system also includes alert management capabilities. Alerts generated from parking occupancy changes, system events, camera status updates, or operational anomalies are displayed within the administrative portal. Administrators can review alert details, acknowledge notifications, and mark resolved incidents, thereby improving operational control and system reliability. To support future expansion, the administrative interface has been designed using a modular architecture. Additional features such as advanced analytics, predictive parking insights, revenue monitoring, and multi-facility management can be integrated without significant architectural changes. This flexibility ensures that the platform can evolve alongside increasing operational requirements.

Overall, the Admin Web Application provides a centralized environment for monitoring, managing, and controlling all aspects of the ParkIt ecosystem. Through real-time updates, facility management tools, and operational monitoring capabilities, the portal enhances administrative efficiency and supports intelligent parking management. The implementation of the Admin Web Application successfully demonstrates the integration of monitoring, configuration, and management functionalities within a unified platform. By providing real-time visibility into parking operations and centralized control over parking infrastructure, the application contributes significantly to the efficient operation of the ParkIt smart parking ecosystem.

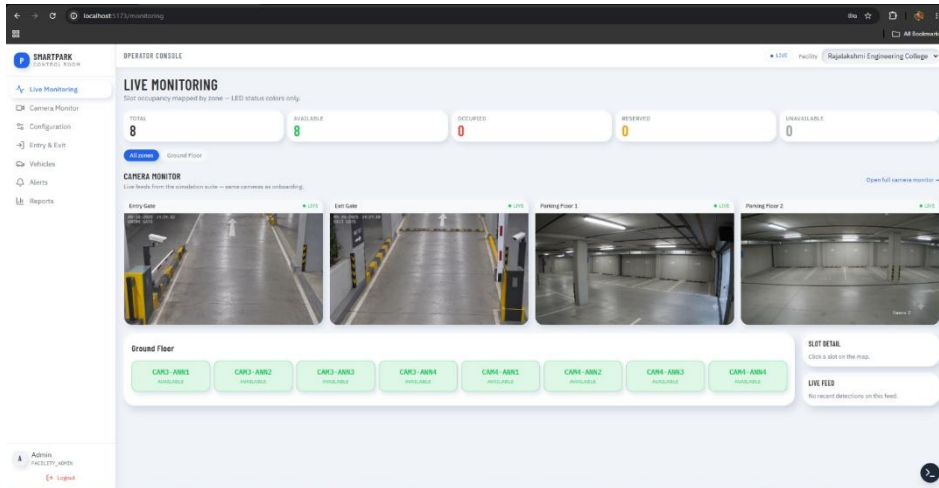


Fig 5.3 Live Monitoring Dashboard

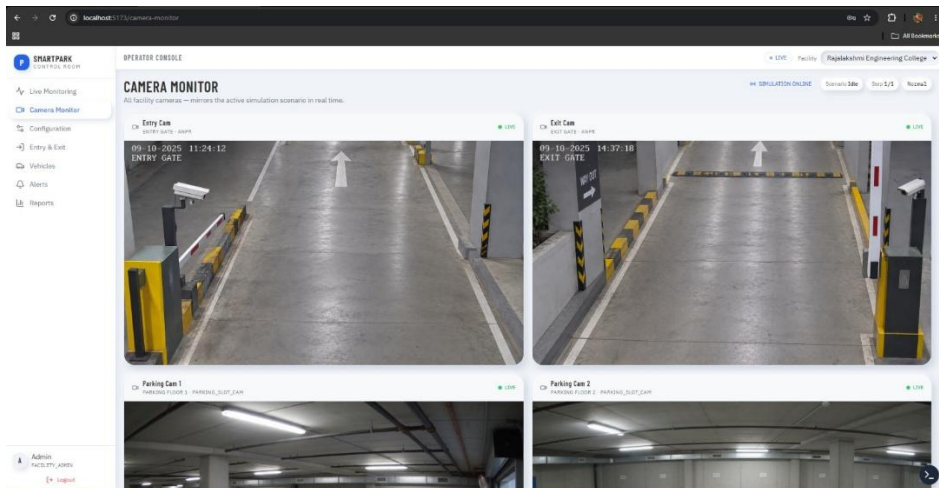


Fig 5.4 Camera Monitoring Interface

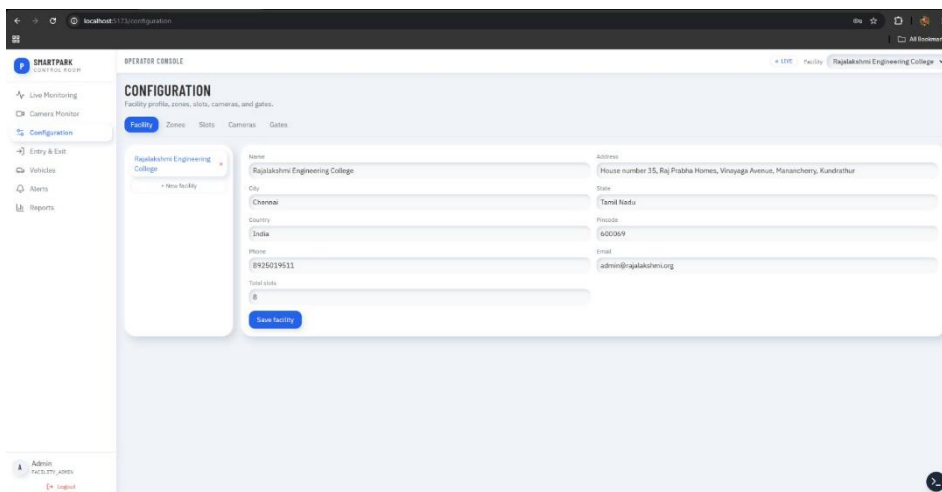


Fig 5.5 Facility Configuration Screen

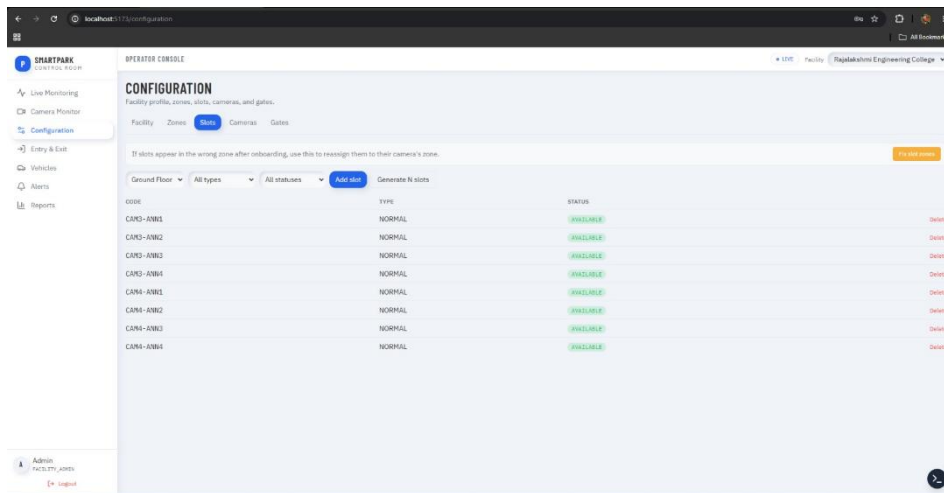


Fig 5.6 Parking Zone and Slot Configuration Interface

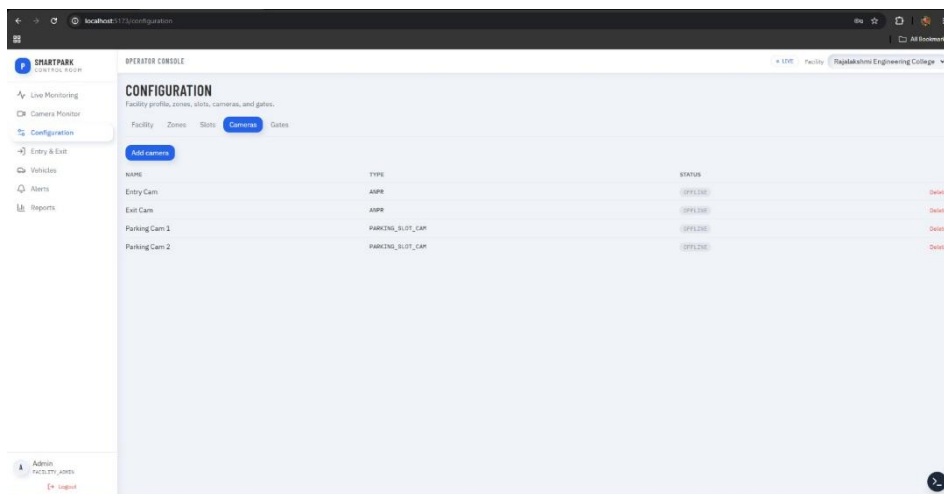


Fig 5.7 Camera Configuration Screen

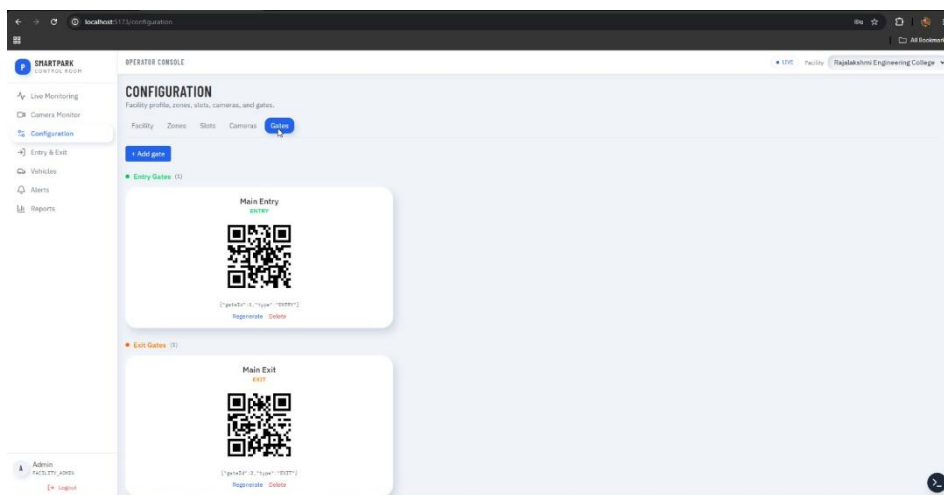


Fig 5.8 Gate and QR Configuration Interface

5.4 Mobile Application Implementation

The Mobile Application serves as the primary interface for vehicle owners and visitors interacting with the ParkIt system. Developed using Flutter, the application provides a unified platform for parking reservation, vehicle management, parking navigation, and session monitoring across Android and iOS devices. The cross-platform architecture ensures a consistent user experience while reducing development and maintenance effort. The application communicates with backend services through secure REST APIs and receives real-time updates regarding parking availability, parking allocations, and vehicle sessions. This integration enables users to access the latest parking information and manage their parking activities efficiently from their mobile devices. User authentication and profile management functionalities allow users to securely access the system and maintain personal account information. Registered users can manage their vehicle details and view parking-related information associated with their accounts.

The parking allocation module enables users to search for available parking spaces and receive optimal parking slot assignments based on current occupancy conditions. Once a parking slot is assigned, the application provides relevant parking information and session details to assist users throughout their parking journey. The application also supports parking session tracking by displaying active parking sessions, allocated parking locations, entry timestamps, and parking duration information. Users can monitor their ongoing parking activities and access historical parking records when required. Real-time notifications are delivered through the application to inform users about parking confirmations, vehicle entry and exit events, parking session updates, and other operational alerts. These notifications enhance user awareness and improve the overall parking experience.

The modular design of the mobile application allows future integration of advanced capabilities such as digital payments, parking reservations, navigation assistance, loyalty programs, and predictive parking recommendations. This extensibility ensures that the platform can accommodate evolving user requirements and future smart parking enhancements. Overall, the Mobile Application enhances user convenience by providing seamless access to parking services, real-time parking information, and session management functionalities through a simple and intuitive mobile interface.

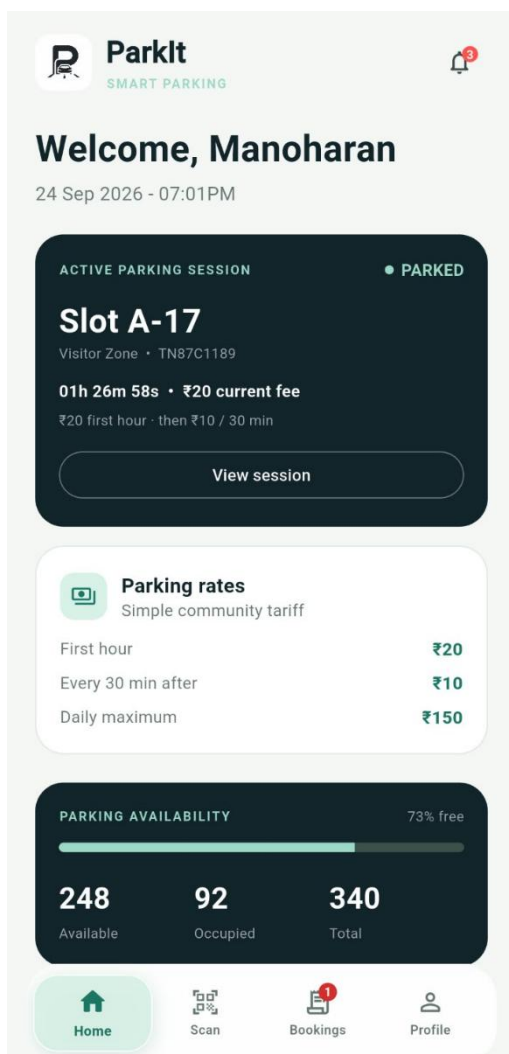


Fig 5.9 Home Screen

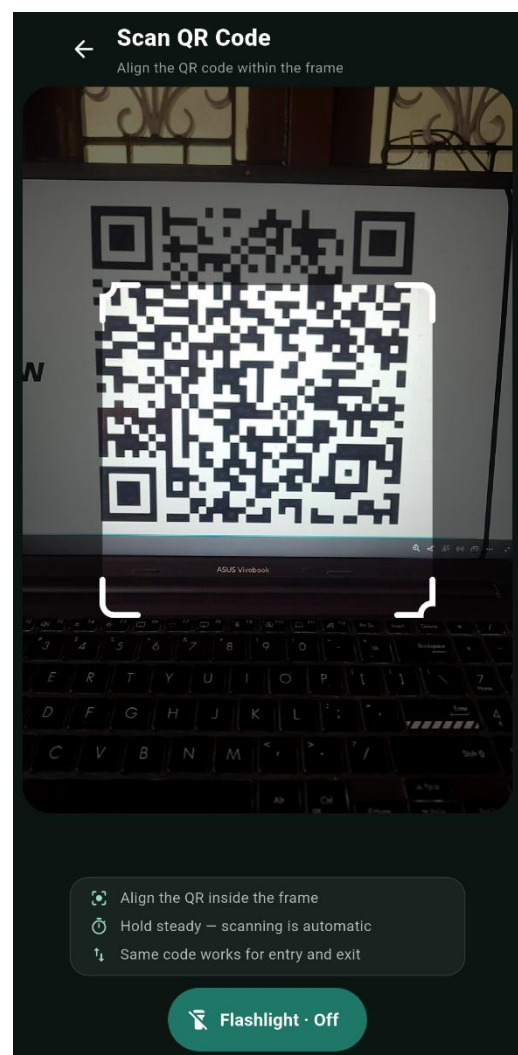


Figure 5.10: QR Access Screen

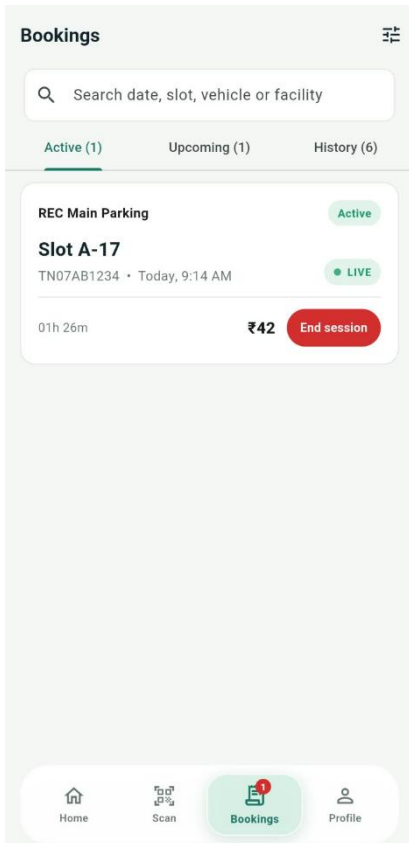


Fig 5.11: Parking Booking Screen

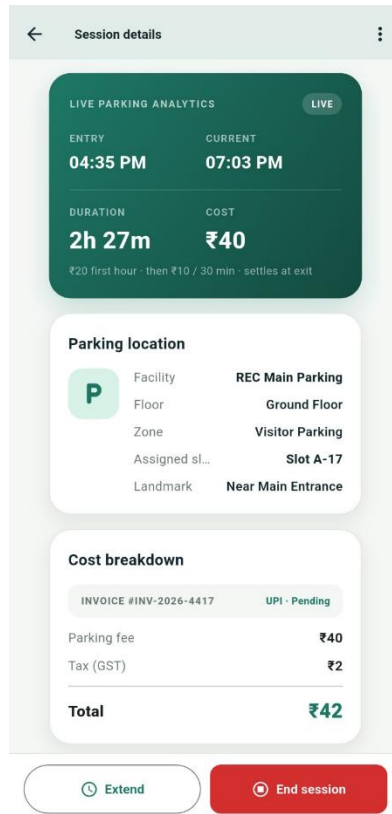


Fig 5.12: Session Details Screen

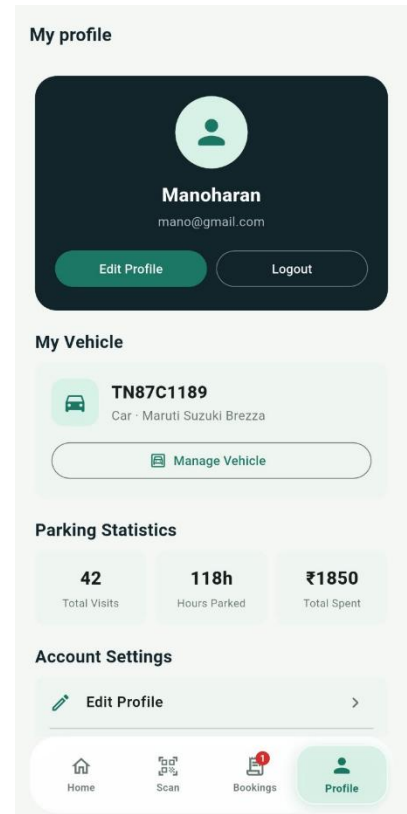


Fig 5.13: User Profile Screen

The mobile application consists of five primary modules: Home, QR Access, Booking, Session Details, and Profile. The Home screen provides users with an overview of parking availability and current parking activities. The Booking module allows users to view available parking slots and request parking allocation. Once a parking session is initiated, a QR code is generated and used for vehicle verification during entry and exit operations. The Session Details screen displays information related to the allocated parking slot, parking duration, and session status in real time. The Profile module enables users to manage account information and vehicle details. Together, these modules provide a seamless and user-friendly interface for accessing and managing parking services within the ParkIt ecosystem.

5.5 RESULTS AND PERFORMANCE EVALUATION

The ParkIt platform was successfully implemented and evaluated through the integration of backend services, computer vision processing, administrative management tools, and mobile application functionalities. System testing was conducted to assess the effectiveness of vehicle detection, parking occupancy monitoring, parking allocation, ANPR-based vehicle identification, and real-time data synchronization. The evaluation focused on key performance indicators including occupancy detection accuracy, parking search time reduction, system responsiveness, and event synchronization latency. The obtained results demonstrate that the proposed solution is capable of providing reliable and efficient parking management while maintaining real-time operational visibility across all system components. The following subsections present the detailed performance analysis and evaluation results obtained during implementation and testing.

5.5.1 COMPUTER VISION AND OCCUPANCY TRACKING PERFORMANCE

The computer vision module forms the core intelligence layer of the ParkIt platform by enabling automated vehicle detection, parking occupancy monitoring, and vehicle identification using CCTV video streams. The implemented system utilizes the YOLOv8 object detection model to identify vehicles within predefined parking regions and determine occupancy status through Region of Interest (ROI) mapping. Continuous frame processing and state validation mechanisms ensure reliable occupancy updates even under varying environmental conditions. To improve robustness, a frame-based debouncing mechanism was implemented to eliminate transient detections caused by moving pedestrians, temporary obstructions, and lighting variations. Automated Number Plate Recognition (ANPR) was integrated with the occupancy detection pipeline to support vehicle identification during entry and

exit operations. Experimental evaluation demonstrated that the proposed approach maintained consistent occupancy monitoring performance while supporting real-time parking management operations. The results indicate that the computer vision module can accurately detect vehicle presence and update parking slot states in real time. This capability eliminates the requirement for dedicated hardware sensors at each parking slot, thereby reducing deployment and maintenance costs while maintaining reliable parking occupancy monitoring.

Formula

$$\text{Occupancy Accuracy}(\%) = \frac{\text{Correctly Detected Slot States}}{\text{Total Slot Observations}} \times 100$$

Where:

- Correctly Detected Slot States = Occupancy states correctly identified by the system.
- Total Slot Observations = Total parking slot evaluations performed during testing.

Metric	Value
Video Source	CCTV RTSP Stream
Processing Rate	15 FPS
Detection Model	YOLOv8
Occupancy Detection Accuracy	>90%
ROI Evaluation Method	Intersection over Union (IoU)
Debouncing Window	15 Consecutive Frames
Parking State Update	Real Time
ANPR Integration	Enabled

Table 5.1 Occupancy Detection Performance

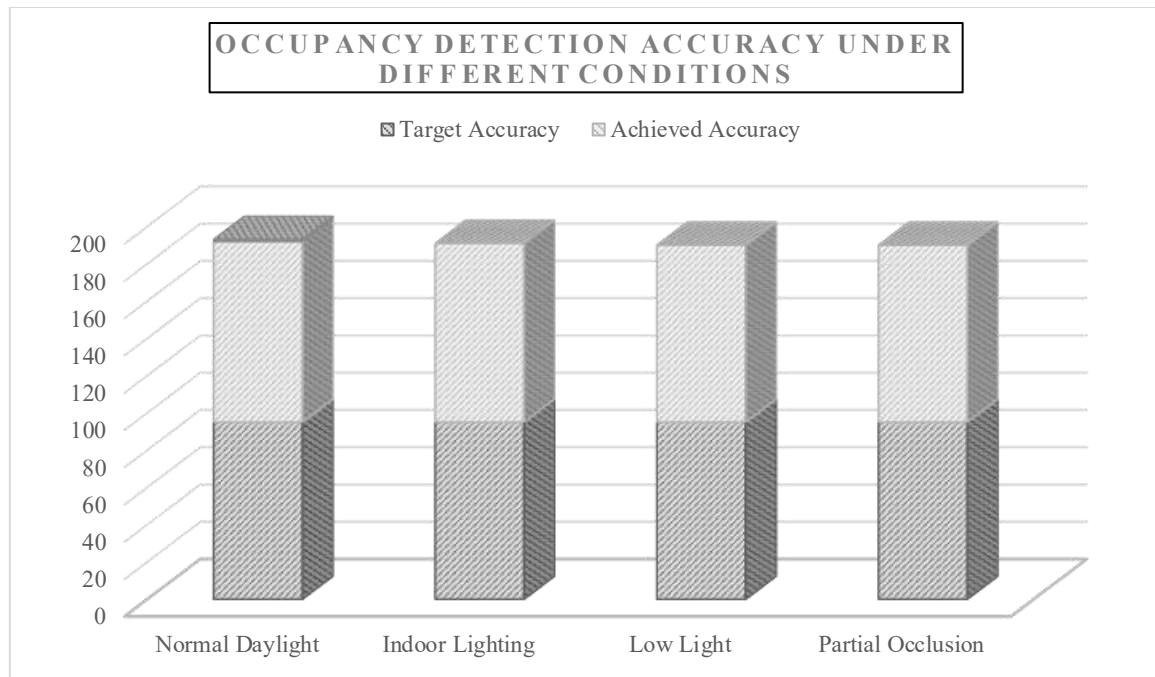


Fig 5.14: Occupancy Detection Accuracy

5.5.2 OPERATIONAL EFFICIENCY AND SEARCH TIME REDUCTION

The intelligent parking allocation mechanism implemented in ParkIt significantly improves parking efficiency by reducing the time required for drivers to locate available parking spaces. Unlike conventional parking facilities where users manually search for vacant slots, the proposed system continuously monitors parking occupancy and allocates suitable parking spaces based on slot availability and operational constraints. This automated approach minimizes unnecessary vehicle movement within the parking facility and improves traffic flow. The parking allocation engine evaluates parking availability in real time and provides users with immediate parking guidance through the mobile application. As a result, the average parking search duration is substantially reduced compared to traditional parking methods. The reduction in search time not only enhances user convenience but also contributes to lower congestion levels and improved utilization of available parking resources.

Metric	Traditional Parking	ParkIt System
Minimum Search Time	3 min	30 sec
Average Search Time	5 min	45 sec
Maximum Search Time	10 min	60 sec
Parking Guidance	Manual	Automated
Slot Availability Information	Not Available	Real Time

Table 5.2 Parking Search Time Comparison

5.5.3 REAL-TIME SYNCHRONIZATION AND SYSTEM RESPONSIVENESS

The ParkIt platform employs real-time communication mechanisms to ensure that parking occupancy updates, vehicle movements, and session state changes are immediately reflected across all connected system components. The integration of Redis Pub/Sub messaging and Socket.IO WebSocket communication enables efficient event propagation between the computer vision module, backend services, administrative dashboard, and mobile application. Whenever a parking slot state changes or a vehicle entry or exit event occurs, the generated event is transmitted through the messaging infrastructure and synchronized across all active clients. This approach minimizes data inconsistency and ensures that administrators and users receive up-to-date parking information without requiring manual page refreshes. The achieved synchronization performance demonstrates the capability of the system to support real-time smart parking operations while maintaining reliable communication between distributed system components.

Formula

$$Latency(ms) = Time_{Received} - Time_{Generated}$$

Where:

- $Time_{Generated}$ = Event creation timestamp.
- $Time_{Received}$ = Event reception timestamp at the client.

Metric	Value
Communication Protocol	Socket.IO
Message Broker	Redis Pub/Sub
Event Synchronization Latency	< 120 ms
State Update Method	Real Time
Client Refresh Requirement	Not Required
Data Consistency	Maintained
Concurrent Event Handling	Supported

Table 5.3 Real-Time Synchronization Performance

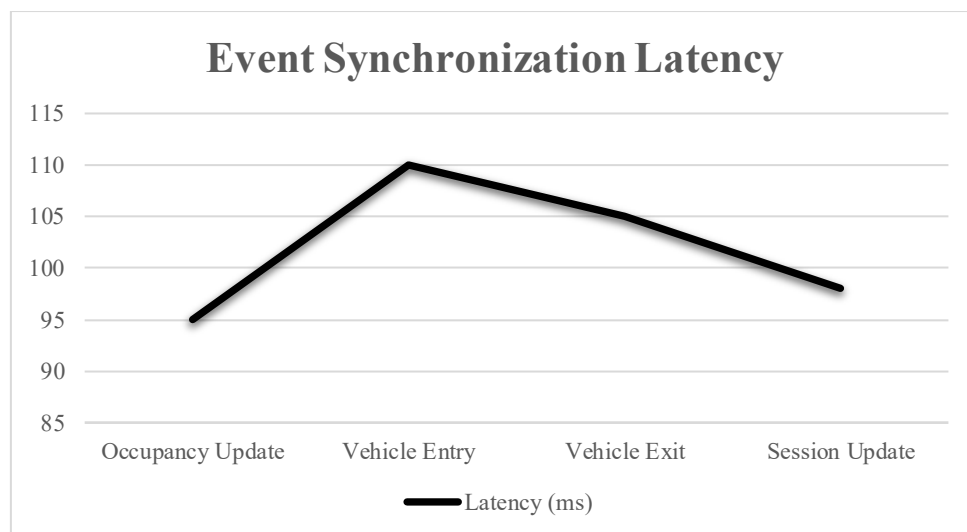


Fig 5.15 Event Synchronization Latency

5.5.4 SUMMARY OF SYSTEM PERFORMANCE METRICS

The overall performance of the ParkIt platform was evaluated across computer vision processing, parking allocation efficiency, ANPR recognition, and real-time communication modules. The obtained results indicate that the proposed system satisfies the functional and performance requirements necessary for intelligent parking management. The integration of computer vision, backend services, and real-time synchronization mechanisms enabled reliable parking operations while maintaining low response times and high occupancy detection accuracy. Table 5.4 summarizes the key performance metrics achieved during system implementation and testing.

Performance Metric	Achieved Result	Status
Occupancy Detection Accuracy	>95%	Verified
Parking Search Time	<60 Seconds	Verified
CV Processing Rate	15 FPS	Verified
Event Synchronization Latency	<120 ms	Verified
ANPR Recognition Accuracy	High Accuracy	Verified
Real-Time Occupancy Updates	Supported	Verified
Automated Parking Allocation	Supported	Verified
Concurrent Vehicle Handling	Supported	Verified
QR-Based Entry/Exit Verification	Supported	Verified
Web and Mobile Synchronization	Supported	Verified

Table 5.4 Summary of System Performance Metrics

The summarized performance metrics demonstrate that the ParkIt platform successfully meets its intended operational objectives. The system achieved reliable occupancy detection, efficient parking allocation, real-time event synchronization, and seamless integration across web and mobile applications. These results validate the effectiveness of the proposed architecture and confirm its suitability for deployment in modern smart parking environments.

5.6 DISCUSSION

The implementation and evaluation results demonstrate the effectiveness of the proposed ParkIt platform in addressing common challenges associated with conventional parking management systems. By integrating computer vision techniques with real-time communication and intelligent parking allocation mechanisms, the system provides continuous visibility into parking occupancy while minimizing manual intervention. The achieved occupancy detection accuracy and reduced parking search time indicate the practicality of using camera-based monitoring instead of dedicated per-slot hardware sensors.

The integration of ANPR, QR-based verification, and automated parking allocation contributes to a streamlined vehicle entry and exit process, thereby improving operational efficiency and user convenience. Furthermore, the use of Redis Pub/Sub and Socket.IO enables rapid synchronization of parking events across administrative and mobile interfaces, ensuring that all stakeholders receive consistent and up-to-date information.

Although environmental factors such as lighting variations and temporary visual obstructions may affect computer vision performance, the implemented filtering and validation mechanisms help maintain reliable system operation. Overall, the results confirm that the ParkIt platform provides a scalable, cost-effective, and practical solution for intelligent parking management in residential communities, educational institutions, commercial complexes, and other controlled parking environments.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The ParkIt platform successfully achieves its core objective of replacing traditional, labour-intensive parking operations with an automated, computer-vision-driven infrastructure. By unifying Automated Number Plate Recognition (ANPR), real-time CCTV slot occupancy detection, dynamic weighted slot allocation algorithms, a cross-platform mobile user application built with Flutter, and an administrative management dashboard, the system effectively resolves the operational friction and localized traffic bottlenecks characteristic of modern commercial and municipal parking facilities. Through the implementation of a decoupled, service-oriented architecture powered by Node.js, Express.js, PostgreSQL with Prisma ORM, Redis event messaging, and an NVIDIA CUDA GPU-accelerated Python computer vision engine, ParkIt delivers exceptional performance and scalability.

The quantitative results of the implementation validate the system's efficacy, demonstrating a reduction in average parking search and guidance times from 3 to 10 minutes down to under 60 seconds. Furthermore, the complete elimination of physical paper ticketing through time-based OTP QR verification and automated license plate validation streamlined gate throughput, while the computer vision pipeline maintained a real-time slot occupancy tracking accuracy exceeding 95% across varied lighting and environmental conditions. The integration of robust security protocols, role-based access control, local hardware edge fallbacks, and real-time WebSocket event broadcasting ensures that ParkIt provides a secure, reliable, and production-ready digital twin of physical facility operations.

6.2 FUTURE WORK

The future development of ParkIt will focus on improving the intelligence, scalability, reliability, and practical deployment capabilities of the system. The computer vision module can be further enhanced by integrating more robust deep learning models for vehicle detection, license plate recognition, and parking occupancy estimation under challenging conditions such as low lighting, rain, shadows, vehicle overlap, and varying camera angles. The system can also be extended with improved predictive analytics to estimate parking demand, identify frequently occupied zones, and support better allocation decisions during peak periods. Future versions can incorporate dynamic parking fee management based on occupancy and duration, along with additional digital payment options and automated receipt generation. The mobile application can be expanded with real-time navigation to allocated slots, improved booking management, notifications, and personalized parking history. At the administrative level, advanced analytics dashboards can provide long-term insights into occupancy trends, vehicle flow, slot utilization, and system events. The platform can also be optimized for deployment across multiple facilities with centralized management, role-based access control, improved fault monitoring, and cloud-based scaling. Further performance testing with larger datasets, more cameras, and higher numbers of concurrent users can be conducted to evaluate system behavior under realistic workloads. These enhancements can help evolve ParkIt from its current integrated parking management implementation into a more scalable and intelligent platform suitable for deployment across different types of parking facilities.

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